

Alexander Spangher

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Objective: Measuring and learning *tacit knowledge* to improve creative discovery, planning, and goal-finding.

EDUCATION **Stanford University**, Postdoc, SAIL and HAI Fellow, C.S. *2025–present*
University of Southern California, Ph.D., Computer Science *2019–2025*
Columbia University, M.S. Data Science, M.S. Journalism *2014–2018*
Columbia University, B.S. in Neuroscience, B.S. in C.S. *2010–2014*

RELEVANT PROFESSIONAL EXPERIENCE **Google Research**, *Student Researcher*, CoreML *Summer 2025*
Automated Taxonomy Generation for hierarchical few-shot reasoning, in support of Gemini. Tested hierarchical reasoning for YouTube personalization and math; induced reasoning in untrained reasoning models, raising scores 15–30%.

EleutherAI, *Researcher* *2022–present*
Aria: pretraining of generative piano models (~1B parameters, LLaMA-3.2 architecture, 8 H200s). Contributed pretraining of the Pythia suite (70M–12B). Research on CFG (below) was incorporated into **OpenAI**’s PaperBench: tinyurl.com/cfg-pb

NASA / Carnegie Mellon University, *Research Assistant* *2018–2019*
Knowledge Graph Construction for NASA Earth Sciences Division. Hierarchical topic modeling over scientific text; visualization and tooling in D3.js.

Microsoft Research, *Research Intern* *Summer 2018*
Large-scale analysis of user-behavior changes in response to Russian Internet Research Agency propaganda, working with **Eric Horvitz**: crowdsourcing, causal modeling.

New York Times, *Data Scientist* *2014–2018*
Personalization: Developed article-recommendation algorithms that impacted ~9 million subscribers: a custom Bayesian extension of LDA (with **Dr. David Blei**) <https://nyti.ms/2ngdYmX>. *Project Feels*: Conceived and built emotion models for articles; raised \$40 million. See <https://nyti.ms/2nZCFEt>

SELECTED PUBLICATIONS **A. Spangher**, S. Huang, A. Haupt, N. D. Goodman, D. Yang, D. E. Ho, S. Koyejo. “Verifiable, Articulate, and Tacit Components of Preference.” *arXiv preprint*, 2026
🏆 indicates paper received an award. 🏆 **A. Spangher**, G. Sanchez, H. Fan, E. Levi, S. Biderman et al. “Stay on Topic with Classifier-Free Guidance (CFG).” *ICML 2024*
👤 indicates paper was selected for an oral presentation. 🏆 👤 **A. Spangher**, N. Peng, S. Gehrmann, M. Dredze. “Do LLMs Plan Like Humans? Comparing Journalistic Coverage with LLMs.” *EMNLP 2024*
🏆 **A. Spangher**, X. Ren, J. May, N. Peng. “NewsEdits: A News Article Revision Dataset and a Novel Document-Level Reasoning Challenge.” *NAACL 2022*
👤 **A. Spangher**, G. Ranade, B. Nushi, A. Fournay, E. Horvitz. “Characterizing Search Engine Traffic to Internet Research Agency Web Properties.” *WWW 2020*
A. Spangher, B. Ustun. “Actionable Recourse in Classification.” *ICML 2018*

ADDITIONAL ACTIVITIES & HONORS **Diversity**: Co-chair and Founder: NYT Young Professionals. *Contributor*: Unpublished Black History: <https://nyti.ms/2KvaGDM>
Honors: USC: 4-year Bloomberg Fellowship. Annenberg Fellowship. *Columbia University*: John Jay Scholar and Dean’s List.
Professional Musician: Double Bassist for critically-acclaimed off-Broadway play.
Media Mentions: *Wired*: tinyurl.com/2cojrr2s. *BuzzFeed*: tinyurl.com/y7mq45j8